

3 — Firmware Flashing & Station Provisioning

Audience: Developer / Technical Installer

Prerequisites: VS Code + PlatformIO extension installed

Overview

Each physical ESP32-S3 board must be:

1. **Configured** with its unique Station ID and a provisioning token
2. **Compiled** to produce the firmware binary
3. **Flashed** onto the board over USB
4. **Verified** via the web dashboard

Prerequisites

- [VS Code](#) with the [PlatformIO](#) extension installed
- Clone the repository: `git clone https://github.com/kbvideo6/pumping-station-IoT-500.git`
- Open the `firmware/` folder in VS Code / PlatformIO

Dependencies (auto-installed by PlatformIO on first build)

Library	Purpose
mandulaj/PZEM-004T-v30 @ ^1.1.2	PZEM-004T Modbus driver
adafruit/Adafruit NeoPixel @ ^1.12.3	Status RGB LED driver
adafruit/Adafruit MAX1704X @ ^1.2.2	MAX17048 I2C Battery Gauge driver
bblanchon/ArduinoJson @ ^7.0.0	JSON serialization

Modem Abstraction: To maximize reliability and error handling, this firmware does **not** use `TinyGSM` or the `Firebase-ESP-Client` library. Instead, it interacts with the SIMCom A7670E modem using **direct AT commands over `HardwareSerial`** and the modem's built-in HTTP client stack.

Step 1 — Create the Station on the Dashboard

Before flashing a board, register the station in the cloud:

1. Log into the web dashboard as **Admin**.
2. Navigate to **Stationen verwalten** (Station Management).
3. Click **Station hinzufügen** (Add Station).
4. Fill in:
 - **Station ID** — e.g. `STATION_002` (must be unique, no spaces)
 - **Name** — e.g. `Pumpstation Nord-West`
 - **Pump Power (kW)** — rated motor power
 - **GPS Coordinates** — leave as `0.0 / 0.0`; the onboard GPS will auto-update the location on first boot
5. Click **Provisionieren**.
6. A long string appears — this is the **Provisioning Token**. **Copy it now** (it is only shown once).

Step 2 — Configure the Firmware

Open `firmware/src/config.h` in VS Code and update these two lines:

```
#define DEFAULT_STATION_ID    "STATION_002"           // ← Station ID you created
#define DEFAULT_DEVICE_TOKEN  "PASTE_TOKEN_HERE"     // ← Provisioning token (step 1)
```

All other settings (Firebase URL, APN, pins, thresholds) are pre-configured as defaults and do not need to change for standard deployments.

Optional — Change Alert Thresholds at Compile Time

Default thresholds (can also be changed later per-station on the dashboard):

```
#define DEFAULT_HIGH_CURRENT_THRESHOLD  18.0  // Amperes – overload alert
#define DEFAULT_LOW_CURRENT_THRESHOLD   2.0   // Amperes – no-load / stopped alert
#define DEFAULT_HIGH_VOLTAGE_THRESHOLD  250.0 // Volts – overvoltage alert
#define DEFAULT_LOW_VOLTAGE_THRESHOLD   200.0 // Volts – undervoltage alert
```

Step 3 — Flash the Board

1. Connect the Waveshare ESP32-S3 board to your computer using a **USB-C cable**.
2. In PlatformIO (VS Code bottom toolbar), click the → **Upload** (right-arrow) button.
3. PlatformIO will:
 - Download/update all library dependencies
 - Compile the firmware

- Flash the binary to the board
4. Look for `SUCCESS` in the terminal output.

Command-Line Alternative

```
cd firmware
pio run --target upload
```

Step 4 — First Boot Sequence

After flashing and powering the board (via USB or DIN-rail PSU):

Time	What Happens	LED
0–5s	Board initializes UART, watchdog, PZEM	Solid ON (White Pulse)
5–30s	Modem powers up, connects to 4G network	Fast Blue Blink (2 Hz)
30–60s	Exchanges provisioning token with Firebase for a permanent device token	Fast Blue Blink (4 Hz)
60s+	Begins uploading live PZEM readings every 30 seconds	Green Heartbeat (1 Hz)

Token exchange is permanent — after the first successful boot, `DEFAULT_DEVICE_TOKEN` is no longer used. The board stores its permanent credentials in NVS (non-volatile flash storage).

Step 5 — Verify on Dashboard

1. Open the web dashboard.
2. Go to the **Dashboard** or **Map** view.
3. The new station should appear within 60 seconds as  **Online**.
4. Click the station name to see its live metrics (voltage, current, power, etc.).
5. Verify that GPS coordinates are updating (may take up to 5 minutes for first satellite fix outdoors).

OTA (Over-the-Air) Updates

The firmware supports OTA updates. Deployed boards check for newer versions, but this feature is marked **Coming Soon** in the current release.

Troubleshooting

Symptom	Likely Cause	Fix
Board stuck at fast blink	Invalid provisioning token	Regenerate token on dashboard and re-flash
No PZEM readings (all zeros or --)	Wiring error	Check GPIO 13/14 cross-connection, confirm PZEM powered
No 4G connection	Inactive SIM or wrong APN	Verify SIM activated, check <code>CELLULAR_APN</code> in <code>config.h</code>
Board not appearing on map	GPS no fix	Move GPS antenna to open sky; indoor fix can take 10+ min
Upload fails in PlatformIO	Wrong USB driver	Install CP2104 or CH340 driver for your OS

Next: [Dashboard User Guide →](#)